



Fig. 3: Types of transistors (Source: ELCINA's ELCOMOS report)

TABLE V
Transistors Market Size by Industry Sectors and Future Growth Trends

Application sector	Transistor market size	Growth rate
Telecommunication	12 per cent	30 per cent
Automotive	5 per cent	25 per cent
Consumer durables	35 per cent	18 per cent
IT and office automation	10 per cent	20 per cent
Lighting	25 per cent	12 per cent
Medical electronics	13 per cent for strategic, medical, industrial, etc combined	17 per cent
Industrial electronics		7 per cent
Strategic electronics		15 per cent

Source: ELCINA's ELCOMOS report

TABLE VI
Key Indian Diodes and Transistor Manufacturers

Name of company	Corporate office in India	Products manufactured
Arihant Electricals	Delhi	Transistors (IGBTs)
BEL	Bengaluru	Small-signal transistors
CDIL	New Delhi	Diodes and transistors
Hind Rectifiers Limited	Mumbai	Diodes and transistors
JB Electro	Mumbai	Diodes and transistors
Naina Semiconductor	Noida	Diodes
Ruttonsha International Rectifier Ltd	Mumbai	Diodes
Semikron Electronics	Mumbai	Diodes, IGBTs
Stanesh Semiconductor	Mumbai	Diodes
Vishay Components India	Pune	Diodes
Watts Electronics	Cochin	Diodes

Non-availability of raw material like moulding compound, silicone wafer and lead frames, high import duties on raw material and competi-

ve component pricing from players in China are cited as the key restraints for the transistor manufacturing industry in India.

The tremendous growth of the electronics industry as well as devel-

opment of newer devices has fuelled the growth of transistors. The Indian export of transistors was around \$3 million in FY 2009-10. The demand is mainly from markets like South America and Africa.

Trends in transistors market

The share of indigenous production as compared to that of imports is expected to increase as the two major manufacturers, namely, CDIL and BEL, are planning to expand their facilities.

Transistors are classified based on the types of majority and minority charge carriers introduced in the silicon. Insulated gate bipolar transistors (IGBTs) and metal-oxide field-effect transistors (MOSFETs) are expected to grow at a faster rate than the other transistors due to their usage in devices like power supply units and mobile handsets, respectively.

The key technology trends that are expected to drive the transistor industry in the next few years are increasing usage of nano-transistors in IC sectors like mobile phones and computers, and advent of lithography-free fabrication. Advent of lithography-free fabrication would aid in fabricating clean and high-graphene devices. This method of fabrication would help in production of transistors that operate in the GHz range. This would be the key driver in RF communication in the next few years.

Looking forward

The availability of engineers (who need to be trained) and low cost of labour are the opportunities for India in the near term, while the absence of domestic back-end production facilities and already built overcapacity for back-end production facilities outside India will continue to act as threats. Incentivising such production houses to relocate to India using incentive/taxing tools is important. Let's hope that the policy makers in India will be able to pass this litmus test. ●

The article is based on major inputs from ELCINA's ELCOMOS report and Praveen Alloorak, director-operations, Vishay Semiconductor India